Chapter 7
Bipolar Transistors

The transistor was invented in 1947 first in form of a point contact transistor, whose emitter and collector were formed by sharp metal wires on a germanium block as base [Bar49, Sho49]. Soon it was clear that the metal semiconductor junctions at the point contacts can be replaced by two closely coupled pn-junctions. The first paper on a bipolar transistor of silicon with diffused emitter and base was published in 1956 [Tan56]. For a bipolar transistor as power switch, the emitter and base are fine interdigidated structures whose distance in the range of 30 μm have to be mastered; the technology for this was available in the 1970s. For a certain time, the bipolar transistor was the most important switching device in power electronics. But already at the end of the 1980s, the IGBT was introduced (see Chap. 10), and the IGBT began to replace the bipolar power transistor. Nowadays power converters are no longer equipped with bipolar transistors. In niche markets, e.g. as line deflection transistors in TVs, they have survived. However, recently, activities to develop bipolar SiC transistors were started.

7.1 Function of the Bipolar Transistor

The bipolar transistor has an npn- or pnp-structure. Therefore, it comprises two subsequent pn-junctions. The power transistor, with the exception of the voltage range below 200 V, always has an npn-structure as schematically shown in Fig. 7.1.

When a positive voltage is applied to the collector C, the pn-junction between base B and collector is biased in blocking direction and the pn-junction between base and emitter E is biased in forward direction. With open base, the electron density is low in the base. The base p-doping is in the range between 10^{16} and 10^{17} cm^{-3}; a density $n_{p0} = n_i^2/p$ in the range of 10^4 cm^{-3} can be achieved from the

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J. Lutz et al., *Semiconductor Power Devices*,
https://doi.org/10.1007/978-3-319-70917-8_7

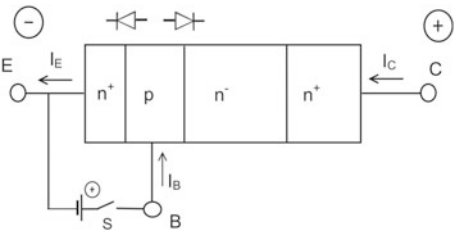


Fig. 7.1 npn power transistor, schematically

relation (2.6). In spite of a high voltage at the collector, the transistor carries only a very small current; it is in the blocking state.

If the switch S is closed and a positive current I_B is injected at the base contact, the n^+p-junction is biased in forward direction, and so the base is flooded with electrons. But with the base current I_B, not only the emitter current is increased. The electrons in the p-base now have a high gradient of carrier density in the direction of the blocking base-collector junction. They diffuse in this direction and into the low-doped n^--layer. If an electric field is build up, they are accelerated by the field towards the collector.

The current gain α in a common-base circuit is defined by

$$I_C = \alpha \cdot I_E + I_{CB0} \tag{7.1}$$

with the leakage current I_{CB0}, measured between base and collector with an open emitter. Furthermore, a current gain β in a common-emitter circuit is defined by

$$I_C = \beta \cdot I_B + I_{CE0} \tag{7.2}$$

with the leakage current I_{CE0}, measured between emitter and collector with an open base. According to Fig. 7.1, I_C is the load current which is to be controlled, therefore β is the current gain of the load current in relation to the controlling base current.[1]

If the relation

$$I_E = I_C + I_B \tag{7.3}$$

resulting from Fig. 7.1 is used and the leakage currents I_{CE0} and I_{BE0} are neglected, Eq. (7.2) is solved for β and finally inserted in Eq. (7.3), it follows that

$$\beta = \frac{I_C}{I_B} = \frac{I_C}{I_E - I_C} = \frac{I_C/I_E}{1 - I_C/I_E} = \frac{\alpha}{1 - \alpha} \tag{7.4}$$

[1]Strictly speaking it must be distinguished between the DC current gain $A = I_C/I_E$ and the small-signal current gains $\alpha = \Delta I_C/\Delta I_E$, the same holds for β. This is neglected in the following simplified treatment.

The same result is also obtained if the exact definitions (7.1) and (7.2) are used; then the leakage currents must be converted. Equation (7.4) can be solved for α which results in

$$\alpha = \frac{\beta}{\beta + 1} \tag{7.5}$$

The closer α approaches 1 the higher is the current gain β of the collector current. For example $\alpha = 0.95$ results in $\beta = 19$.

7.2 Structure of the Bipolar Power Transistor

Figure 7.2 shows the structure of a power transistor. The emitter regions are mostly arranged in stripes, the width of the emitter fingers of power transistors is typically in the range of 200 µm. Base and emitter fingers are arranged in an alternating sequence, interleaved like the teeth of two combs.

The collector region is divided into a low-doped n^--layer to take over the electric field and an adjacent higher-doped n^+-layer. The diffusion profile of the bipolar transistor along the vertical line A–B through an emitter region in Fig. 7.2 is shown in Fig. 7.3. A diffusion profile of this type features the "triple-diffused" bipolar power transistor. The term "triple-diffused" stands for the subsequent diffusion of the deep collector layer, followed by the p-base layer and finally by the n^+-emitter diffusion. In this case, the n^+-layer is a deep-diffused layer with a Gauss-shaped profile of the doping atoms. This deep-diffused layer can also be replaced by an epitaxial layer; then a transistor fabricated form epitaxial wafers is given. But the abrupt junction from the n^+-layer to the n^--layer in epitaxial transistors leads to some disadvantages; see later paragraphs on second breakdown.

7.3 I–V Characteristic of the Power Transistor

In Fig. 7.4, a measurement of the forward I–V characteristic of a power transistor is shown. Already at a low collector voltage, e.g. at 0.4 V, a comparatively high current density is reached. This is not possible for a device where a pn-junction is forward-biased, because of the junction voltage in the range of 0.7 V. In a bipolar transistor in this operation mode, both pn-junctions are forward-biased, and the voltage at the pn^--junction is opposed to that at the n^+p-junction. This region of the forward characteristics, which features very low voltage drop, is denoted as saturation region.

The quasi-saturation region, in which the current slightly increases with increasing voltage, follows next to the saturation region. For a higher voltage – not shown in Fig. 7.4 – the bipolar transistor enters the active region. There, the

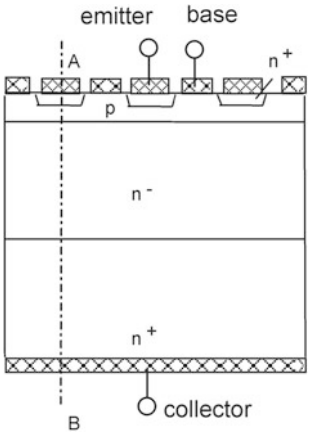


Fig. 7.2 Structure of a power transistor

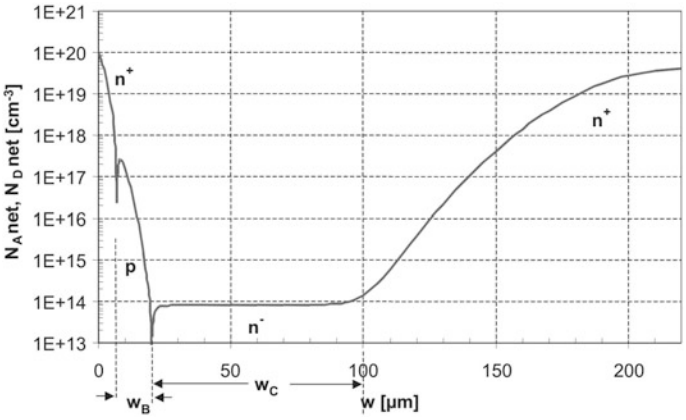


Fig. 7.3 Doping profile of a 1200 V bipolar power transistor along the line A–B in Fig. 7.2

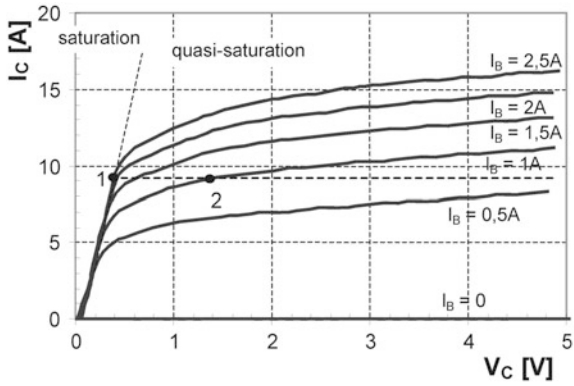


Fig. 7.4 Forward I–V characteristic of a bipolar transistor type BUX 48A

collector current is nearly constant for a given base current and independent of the increasing collector voltage.

From this shape of the forward characteristic, the short circuit capability of the transistor results. The current is limited, even if there is a short circuit at the load. If in a basic circuit according to Fig. 5.18 a short circuit occurs shorting the load consisting of R and L, then the voltage across the transistor increases until the applied external voltage V_{bat} drops across the transistor. The value of the short-circuit current which occurs in short circuit mode is given by the transistor I–V characteristic and the applied base current. Very high losses are generated in this operation mode, however if the short circuit is detected within some µs by appropriate supervising functions in the driver circuit, and is turned-off by the driver, the device will survive this event.

7.4 Blocking Behavior of the Bipolar Power Transistor

In Eq. (7.1), I_{CB0} is the leakage current measured between base and collector. To determine the leakage current between collector and emitter at open base, Eq. (7.1) can be used. At open base, $I_C = I_E = I_{CE0}$ holds, and from (7.1) results that

$$I_{CE0} = \alpha \cdot I_{CE0} + I_{CB0} \tag{7.6}$$

solved for I_{CE0}:

$$I_{CE0} = \frac{I_{CB0}}{1 - \alpha} \tag{7.7}$$

The leakage current between collector and emitter is therefore always higher than the leakage current between collector and base. With $\alpha = 0.9$, I_{CE0} is ten times larger than I_{CB0}.

Also the onset value of avalanche breakdown for a voltage between collector and emitter is lower at open base compared to a voltage between collector and base. To calculate this, Eq. (7.1) must be extended by the effect of avalanche multiplication. The current αI_E enters the space charge from the collector side and is enhanced by the electron multiplication factor M_n. The leakage current I_{CB0}, mainly created by generation in the space charge, is enhanced by its multiplication factor M_{SC} [compare Sect. 3.3 on avalanche breakdown, especially Eq. (3.68)]. It holds that

$$I_C = M_n \cdot \alpha \cdot I_E + M_{SC} \cdot I_{CB0} \tag{7.8}$$

At open base, $I_C = I_E$ is given, which for the collector current results in

$$I_C = \frac{M_{SC} \cdot I_{CB0}}{1 - M_n \cdot \alpha} \tag{7.9}$$

Therefore, the collector current grows to infinity for $M_n \cdot \alpha = 1$ or it holds

$$M_n = \frac{1}{\alpha} \tag{7.10}$$

for avalanche breakdown between collector and emitter, whereas the avalanche breakdown between collector and base arises only if M_n grows to infinity. For high current gain (α close to 1), the blocking capability at open base is strongly reduced. For $\alpha = 0.9$, it is sufficient as the condition for avalanche breakdown that the multiplication factor M_n grows to 1/0.9 = 1.11.

Because the ionization factors for electrons are much higher than that for holes, $\alpha_n > \alpha_p$, M_n grows much faster with the electric field (see Fig. 3.15). M_n determines the avalanche breakdown and the approximation by an effective multiplication factor M or an effective ionization rate α_{eff} is no longer valid. The breakdown voltage V_{CE0} is considerably smaller than V_{CB0}. Figure 7.5 shows the difference between V_{CB0} and V_{CE0} for a commercially available bipolar transistor. For this example, V_{CE0} amounts about 60% of V_{CB0}.

For a pnp-transistor, we obtain in analogy

$$M_p = \frac{1}{\alpha} \tag{7.11}$$

Because of $M_p < M_n$, the difference from V_{CE0} to V_{CB0} will be smaller for a pnp-transistor.

In Sect. 3.3, with Eq. (3.69) an often used approximation for the multiplication factor at a voltage $V < V_{CE0}$ was given [Mil85]

$$M = \frac{1}{1 - (V/V_{CB0})^m} \tag{7.12}$$

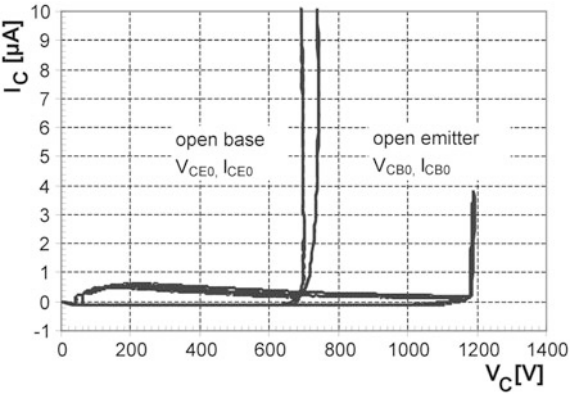


Fig. 7.5 Blocking characteristic of the transistor BUX 48A with open base and with open emitter

where V is an applied voltage smaller than V_{CE0}. With (7.10), it leads for $V = V_{CE0}$ to

$$V_{CE0} = (1 - \alpha)^{\frac{1}{m}} \cdot V_{CB0} \tag{7.13}$$

where a value of 5 for m as is given as typical for a bipolar npn-transistor [Ben99]. The value $m = 5$ leads to agreement with measurements, if for α the value at typical forward operation conditions is used where both β and α are high. However, static avalanche breakdown effects appear at low current where the current amplification is low, since a high share of the current recombines in the base layer. If this is considered, $m = 2.2$ as suggested in Sect. 3.3 is a better choice.

An approximation for a transistor with typical structure can be found also for V_{CE0}, [Ben99]

$$V_{CE0} = K_1 \cdot w_C \tag{7.15}$$

where w_C is the width of the low-doped collector zone, as defined in Fig. 7.3, and K_1 is given as 10^5 V/cm.

The difference between reverse blocking voltage and reverse current of a transistor structure with open base and the reverse blocking voltage of a pure pn-junction is very important for devices with several pn-junctions. It is also very fundamental in the practical application of the bipolar transistor, since for a voltage applied at open base, the device goes into breakdown at a much lower voltage and it is potentially destroyed. On the other hand, if a negative voltage is applied to the base with respect to the emitter, both pn-junctions of the transistor are biased in reverse direction. The leakage current of both junctions is extracted as base current, and no more interaction of both pn-junctions occurs. For this case, the blocking capability between collector and emitter is approximately the same as between collector and base. A similar effect occurs if base and collector terminal are connected by a short circuit. In practical application, a small negative voltage is applied at the base of the transistor if a high reverse voltage emerges at the collector. Usually a negative voltage at the base is applied for turning off the collector current, and it is maintained continuously during the blocking mode.

7.5 Current Gain of the Bipolar Transistor

According to definition (7.1), it holds that

$$\alpha = \frac{j_C - j_{CB0}}{j_E} \tag{7.16}$$

Equation (7.16) is multiplied in nominator and denominator by the electron current j_{nB} which is injected by the emitter into the base

$$\alpha = \frac{j_{nB}}{j_E} \cdot \frac{j_C - j_{CB0}}{j_{nB}} = \gamma \cdot \alpha_T \tag{7.17}$$

The first term on the right-hand side of Eq. (7.17) corresponds to the emitter efficiency γ, which was already introduced in Chap. 3.4 with Eq. (3.99) and which is given for an n-emitter by

$$\gamma = \frac{j_{nB}}{j_{nE} + j_{pE}} \tag{7.18}$$

For an n-emitter, this definition describes the share of the electron current j_{nB}, injected into the base, related to the total emitter current.

The second term in (7.17) is denoted as transport factor α_T

$$\alpha_T = \frac{j_C - j_{CB0}}{j_{nB}} \tag{7.19}$$

For an npn-transistor, this corresponds to the share of the electron current injected by the emitter which arrives at the collector. For $j_C = j_{CB0}$ and therefore $\alpha_T = 0$, only the leakage current arrives at the emitter. For an npn-transistor with high current gain, γ as well as α_T shall be close to unity, so that α exhibits a value close to one.

Now the emitter efficiency γ shall be investigated in more detail. Recombination at the pn-junction between emitter and base shall be neglected; this is feasible at current densities higher than 1 mA/cm^2. The electron currents on both sides of this pn-junction are therefore assumed to be equal, $j_{nE} = j_{nB}$. Then it follows that

$$\gamma = \frac{j_{nB}}{j_{nE} + j_{pE}} = \frac{1}{1 + \frac{j_{pE}}{j_{nB}}} \tag{7.20}$$

The minority carrier current j_{pE} penetrating into the emitter can be expressed by

$$j_{pE} = q \cdot \frac{D_p}{L_p \cdot N_E} \tag{7.21}$$

and for the electron current j_{nB} penetrating into the base at the condition of low injection it holds

$$j_{nB} = q \cdot \frac{D_n}{L_n \cdot N_B} \tag{7.22}$$

Equations (7.21) and (7.22) are inserted in Eq. (7.20). For the case of low injection, that means that the flooding with free carriers is smaller than the doping of the base N_B, the emitter efficiency γ can now be expressed by

$$\gamma = \frac{1}{1 + \frac{D_p}{D_n} \cdot \frac{N_B}{N_E} \cdot \frac{w_B}{L_p}} \tag{7.23}$$

In this equation L_n, the diffusion length of the electrons in the base, was replaced by the width of the base w_B, since w_B is always smaller than L_n for the high carrier lifetime typical in bipolar transistors. Further L_p in the emitter will be small, after all w_B and L_p will be in the same order or magnitude. Therefore, the dominating term in (7.23) is the quotient N_B/N_E. To achieve γ close to one, the doping of the emitter N_E must be much higher than the doping of the base N_B. Equation (7.23) gives a first approximation for these relations, which are very important for the design of a transistor.

Equation (7.23) is valid for the case of low injection. Furthermore, bandgap narrowing was not considered in Eq. (7.23); that means a not too high doping of the n^+-emitter was presumed. Additionally, the description of the emitter efficiency in Eq. (7.23) does not contain the current dependency of the emitter efficiency. Based on Sect. 3.4, the emitter efficiency can be investigated in more detail. The n-emitter is characterized by the emitter parameter h_n. Auger recombination and bandgap narrowing determine an n-emitter with high doping; in analogy to (3.108) it holds that

$$h_n = e^{\Delta E_g / kT} \cdot \sqrt{D_p \cdot c_{A,n}} \tag{7.23a}$$

With the Auger coefficient $C_{A,p} = 2.8 \times 10^{-31}$ cm^6/s and with the mobility μ_p of 79 cm^2/Vs (estimated for a doping of 1×10^{19}cm^{-3}) and using bandgap narrowing according to Slotboom and DeGraaf as described in Eq. (2.25) one obtains

$$h_n \approx 2 \times 10^{-14} \mathrm{cm}^4/s$$

The experience of the authors with design and fabrication of bipolar transistors has shown that the parameter h_n is in the range between 1×10^{-14} and 2×10^{-14} cm^4/s up to a doping of 5×10^{19} cm^{-3}, and it only increases at higher doping as results from (7.23a) with bandgap narrowing according to (2.25). A high value for h_n indicates a reduced emitter efficiency.

The emitter efficiency can be expressed in analogy to (3.100) as

$$\gamma = 1 - q \cdot h_n \frac{p_L^2}{j} \tag{7.24}$$

To estimate γ, the density of the free carriers p_L at the junction between emitter and p-base must be known. In a power transistor, shown again schematically in Fig. 7.6 (top), a low-doped collector layer with a thickness w_C follows adjacent to the p-base. This layer is responsible for taking over the space charge in the forward blocking mode of the transistor. It must be flooded with free carriers in on-state to achieve a low-voltage drop in the conduction mode. The effective base width increases from w_B to $w_B + w_C$ (Fig. 7.6), referred as Kirk-effect [Kir62].

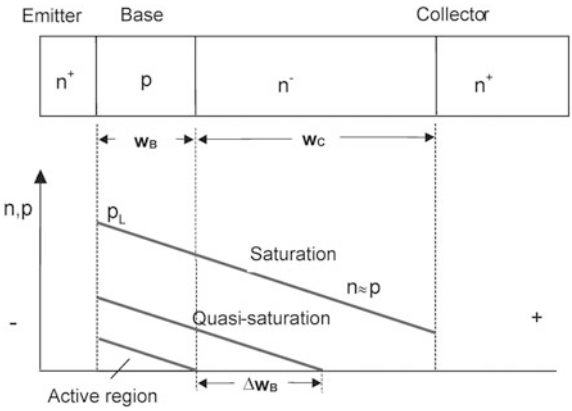


Fig. 7.6 Density of the plasma of free carriers in a bipolar transistor for varied base current at constant collector current

The transistor is in the mode of saturation at high base current and low voltage V_C. This point is marked in the I–V characteristic in Fig. 7.4 with the number 1. The holes injected in the base diffuse also into the low-doped collector region. A conductivity-modulated zone is built up in which $n \approx p$ holds. The shape of the carrier density at point 1 in the I–V characteristic is shown in Fig. 7.6 with the graph for the mode of saturation. The p-base as well as the low-doped collector layer are flooded with free charge carriers. The current transport from the collector to the emitter can be considered as solely carried by electrons, since the diffusion of holes is oriented in the direction against the electric field, whereas the diffusion of electrons is supported by the electric field.

The density of free carriers decreases from the emitter towards the collector. For the distribution across base and low-doped collector layer, Eq. (5.25) still applies. This equation leads to a sagging carrier distribution. The grade of deviation from a linear shape is determined by the charge carrier lifetime. It turns out to be more pronounced for a lower carrier lifetime. In high-quality bipolar transistors, the carrier lifetime is high and the loss by recombination is low, therefore this sagging is neglected in Fig. 7.6. This is a good approximation for $L_n > 2 \cdot (w_B + w_C)$.

Therefore, with the assumed simplified relation

$$\frac{dp}{dx} = -\frac{p_L}{w_B + w_C} \tag{7.25}$$

which applies for the point of transition from the saturation mode to the quasi-saturation mode, see Fig. 7.6, and with $j_p = 0$, $j = j_C$, the result obtained for the collector current is based on Eq. (5.21).

$$j_C = \frac{\mu_n + \mu_p}{\mu_p} \cdot q \cdot D_A \cdot \frac{p_L}{w_B + w_C} \tag{7.26}$$

Using Eq. (5.23) and the Einstein relations (2.44),

$$j_C = 2 \cdot q \cdot D_n \cdot \frac{p_L}{w_B + w_C} \tag{7.27}$$

is obtained. Solved for the density of free carriers at the emitter-base junction

$$p_L = \frac{j_C \cdot (w_B + w_C)}{2D_n \cdot q} \tag{7.28}$$

follows. Equation (7.28) inserted in (7.24) now allows the estimation of the emitter efficiency. For a transistor with w_C = 50 µm for example and w_B = 10 µm, a value for p_L of 2×10^{16} cm^{-3} results for a current density of 30 A/cm^2. The emitter efficiency is γ = 0.96. For 10 A/cm^2, a value of γ = 0.99 is obtained. The emitter efficiency depends strongly on the current density. With increasing collector current, the emitter efficiency and thereby the current gain decreases. This is observed for all power transistors (see Fig. 7.7 for high I_C).

The transport factor α_T was the second factor for the current gain according to Eq. (7.17), it can be written for $L_n > 2 \cdot w_{eff}$ [Sze81, Ben99]

$$\alpha_T = 1 - \frac{w_{eff}^2}{2 \cdot L_n^2} \tag{7.29}$$

where L_n represents the diffusion length in the base, which is combined with the carrier lifetime according to Eq. (3.48). w_{eff} is the effective width of the base, which may be shortened compared to w_B at increased voltage, see Fig. 7.8. Equation (7.29) is usually derived for the low-injection condition, for details see [Ben99]. To attain α_T close to 1, w_B must be chosen small and L_n must be the maximum possible. A high current gain requires a short base and a carrier lifetime in the base as high as possible.

The current gain of the bipolar transistor depends on the current and the temperature. For the current gain β, this is shown in Fig. 7.7.

For very small current, β is small; the current injected into the base recombines to a large extent in the base layer. For increasing current, β reaches a maximum; for further increasing current, at the condition of high injection, it decreases again. This

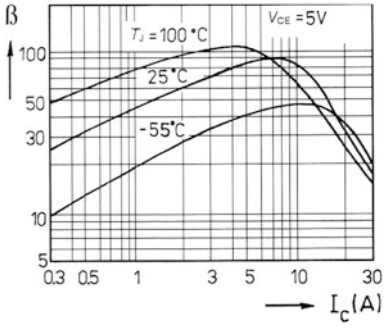


Fig. 7.7 Current gain β as function of collector current and temperature. Data by M. Otsuka, Development of High Power Transistors for Power Use, Toshiba 1975. Reprint from [Ben99] with permission of John Wiley & Sons, Inc

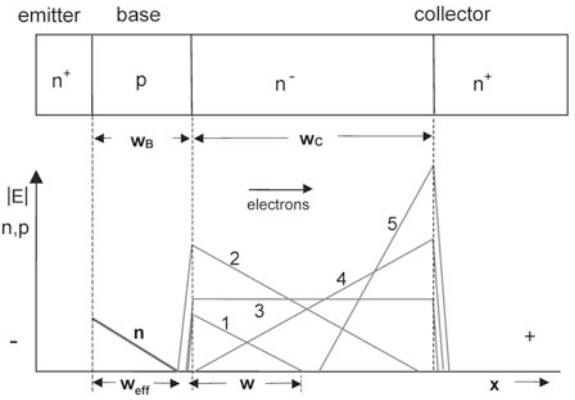


Fig. 7.8 Electric field in the active mode of the bipolar transistor. 1 → 2: increasing voltage V_{CE}. 2 → 3, 4, 5: constant voltage, increasing current I_C

is attributed to the emitter efficiency, which decreases as described by Eqs. (7.24) and (7.28). Additionally, the temperature dependency must be taken into consideration. β increases with temperature for small and medium current, because the carrier lifetime increases with temperature. For high current, the decrease of β with temperature is shifted to lower currents. The rated current of a bipolar transistor is typically in the range where β und α are already declining.

7.6 Base Widening, Field Redistribution and Second Breakdown

If the base current is reduced while the collector current remains constant, the transistor transits into the mode of quasi-saturation; in the I–V characteristic in Fig. 7.4, this is the shift from point 1 to point 2. Now the low-doped collector layer is flooded with plasma of free carriers only up to Δw_B in Fig. 7.6. Only electrons carry the current in the region w_C–Δw_B which is free of plasma, and because of the low doping, a significant resistive voltage drop is created. It can be described in analogy to Eq. (6.8) by

$$\Delta V_{CE} = \frac{j_C \cdot (w_C - \Delta w_B)}{q \cdot \mu_n \cdot N_D} \tag{7.30}$$

If the collector current is kept constant and the base current is decreased, then Δw_B decreases as shown in Fig. 7.6. For $\Delta w_B = 0$, only the base w_B is flooded with free carriers; the transistor has now reached the active region. At this point, $\Delta V_{CE} = 13.4$ V results for the example of a 600-V-transistor with $w_C = 60$ µm, background doping $N_D = 1 \times 10^{14}$ cm^{-3}, $j_C = 50$ A/cm^2 and $\mu_n = 1400$ cm^2/Vs.

Now the pn-junction between base and collector has become free of carriers. If now the voltage is increased, a space charge will build up. For a moderate voltage this is shown in Fig. 7.8 with line 1, for increased voltage line 2. The flooded zone

is pushed back further into the base by the amount that the space charge penetrates the higher doped base layer. The base width is shortened to the effective base width w_{eff}, which leads to a slight increase in the current gain. In literature, this is described as Early effect (named after its discoverer James M. Early, 1922–2004) [Ear52]. Power transistors are usually not operated in the active region. However, they pass through the active region at switching instants.

For a voltage in the active region close to the voltage V_{CE0}, the electric field is shown in Fig. 7.8, line 2. The space charge has build up over nearly the whole layer w_C for this case.

The collector current now flows through the space charge as electron current. The condition of a high electric field applies, and the electrons travel with the drift velocity $v_d \approx v_{sat}$ over almost the whole distance. For the electron density holds

$$n = \frac{j}{q \cdot v_{sat}} \tag{7.31}$$

The shape of line 2 in Fig. 7.8 is given only as long as the density of electrons travelling through the n^--layer is small compared to the background doping N_D. The negatively charged electrons compensate the background doping, and according to Poisson's equation it is

$$\frac{dE}{dx} = \frac{q}{\varepsilon}(N_D - n) \tag{7.32}$$

If the collector current increases by increased base current, the condition will be reached that the density of electrons flowing through the space charge is equal to the background doping:

$$\frac{j}{q \cdot v_{sat}} = N_D \tag{7.33}$$

For this case, $dE/dx = 0$ applies and an almost rectangular electric field with the shape of line 3 in Fig. 7.8 exists. For a bipolar transistor with the background doping $N_D = 1 \times 10^{14}$ cm^{-3}, Eq. (7.33) is fulfilled at a current density of approximately 160 A/cm^2, using $v_{sat} = 10^7$ cm/s.

If the collector is current further increased by an increased base current, then $n > N_D$ holds, and the gradient of the electric field changes its sign, see line 4 in Fig. 7.8 [Hwa70]. The electric field is redistributed and the field maximum shifts from the pn-junction to the nn^+-junction. A further increase in the current leads to a further increased field strength (line 5), and finally avalanche breakdown at the nn^+-junction will occur.

This is the onset of second breakdown. This effect was explained first by Phil Hower [How70]. Second breakdown is destructive: holes, generated by avalanche at the nn^+-junction, are accelerated in the layer w_C. The front side of the transistor is in the active mode, the arriving holes act as an additional base current and even

more electrons are generated by the emitter etc. A positive feedback loop is established. Such a mechanism is usually destructive.

To avoid this critical condition, a safe operating area (SOA) is defined for the transistor for the case of positively biased base – FBSOA, forward-biased SOA – and for the case of negatively biased base – RBSOA, reverse biased SOA. The turn-off is a particularly critical condition in this respect. During turn-off with inductive load, the voltage must increase first before the current can decrease. The transistor passes through the active region of the I–V characteristic. The specification of an RBSOA sets a limit for the voltage against which the transistor can be turned off. Figure 7.9 shows an example.

The turn-off process is also critical because of the following reason: the current below an emitter finger is extracted from both edges toward the center. At turn-off with an inductive load, a small area in the center of the emitter finger finally remains, which carries the total current. Thus, the current density is increased and the mechanism according to Eqs. (7.31) – (7.33) is triggered at lower currents. Some design measures are possible to increase the SOA. The width of the emitter fingers can be reduced and the pitch of the structure can be diminished. Also special emitter structures were tested, which maintain the current flow at the edge of the emitter, e.g. the ring emitter structure [Mil85]. These structures showed an extended RBSOA and a considerably improved stability against second breakdown.

Finally, a shallow gradient of the diffusion profile at the nn^+-junction, as shown in Fig. 7.3, is a countermeasure against the formation of a field peak at the nn^+-junction. For a slightly increasing n-doping towards the collector, the electric field can penetrate significantly into the n collector layer and a higher voltage can be sustained before the conditions for avalanche breakdown are encountered. Usable transistors in the range of 1000–1400 V could be manufactured with a diffusion profile similar to Fig. 7.3.

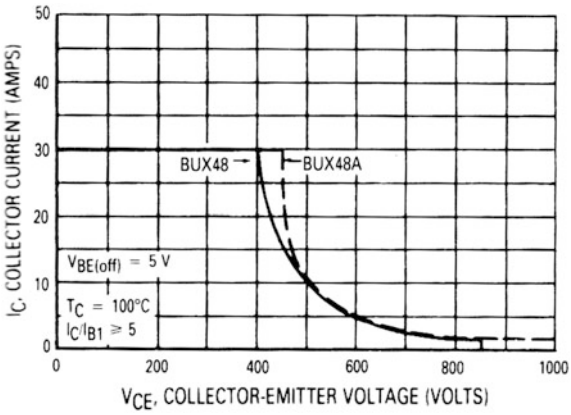


Fig. 7.9 RBSOA of the Motorola transistor BUX 48

7.7 Limits of the Silicon Bipolar Transistor

If the transistor is designed for higher voltage, the low-doped collector layer w_C must be widened. Since the function of the bipolar transistor is based on diffusion of holes into this low-doped collector region, the current gain decreases with increasing w_C. The Motorola transistor in Fig. 7.4 exhibits $\beta = 10$ only up to a collector current of 10 A. The high base current requires a high effort and generates significant losses in the base drive units.

The required base current could be reduced down to acceptable values by the introduction of double- and triple-stepped Darlington transistors [Whe76]. Darlington transistors with blocking voltages of 1200–1400 V became available with up to 100 A of controllable current per single die. With Darlington transistors, high-switching frequencies are no longer possible. But the requirements for variable speed motor drives with switching frequencies in the range of 5 kHz could be fulfilled by Darlington transistors.

An extension to higher voltages is not possible with silicon. Meanwhile, a field-controlled device, the IGBT, was found for the motor drive applications. IGBTs are much easier to control and cause lower losses in the driver. Therefore, bipolar power transistors have been widely taken from the market of power devices and replaced by IGBTs. But the knowledge of the effects in bipolar transistors is essential for a deeper understanding of the effects in more complex power devices.

7.8 SiC Bipolar Transistor

With SiC, a bipolar transistor with a much thinner collector layer and therefore a drastically reduced w_C is possible. Figure 6.8, which determines the necessary width of the low doped region, is also valid for dimensioning a SiC bipolar transistor. According to the small w_C, an acceptable current gain can be achieved even for transistors with blocking voltages above 1000 V. To obtain a high current gain, state-of-the-art SiC epitaxial growth and surface passivation are important. SiC bipolar transistors can be fabricated with very low on-state voltage drop, if ohmic contacts with low contact resistivity are formed [Lee07].

Figure 7.10 shows the measurement of a SiC "large area" bipolar transistor, fabricated by TranSiC AB [Dom09]. A value for β of 35 was achieved for a BJT with open-base breakdown voltage V_{CE0} of 2.3 kV. It should be noted that the characteristic is nearly ohmic in the saturation mode, and an "on-resistance" of 0.03 Ω can be observed, which is about 0.45 Ω mm^2 for this device with 15 mm^2 active area.

The SiC transistor establishes the opportunity to operate a high-voltage device with a voltage drop clearly below 1 V at room temperature. Further, SiC has the advantage of a possible higher doping of the region w_C. As to be see in Fig. 3.19, the maximum doping concentration for a given blocking voltage is for SiC two

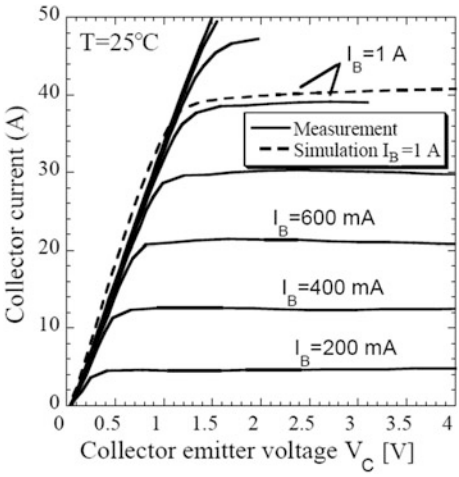


Fig. 7.10 I–V characteristic of a SiC bipolar transistor. Active area 15 mm^2, breakdown voltage V_{CE0} = 2.3 kV

decades higher than for Si. Therefore, the effect of second breakdown according to Eqs. (7.31)–(7.33) is to be expected only at a very high current density which is outside the range of possible operation. For 1200 V SiC BJT, successful operation at V_C = 1100 V and a current density up to 2990 A/cm^2 has been shown by measurement in [Gao07]. In contrast to the Si BJT, the SiC BJT is not restricted by the second breakdown in the application-relevant area.

Additionally, high operation temperatures are possible with SiC, but then a reduced current gain and an increased on-resistance must be taken into account. The progress in SiC technology had revitalized the interest in bipolar transistors again. However, in power device application the SiC bipolar transistor competes with the SiC MOSFET.

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